

KASCC 2026 — ESP8266 IOT PROJECTS

KASCC 2026 — ESP8266 IoT Projects Collection

A collection of 7 hands-on IoT projects built with the ESP8266 (NodeMCU) microcontroller. Each project demonstrates real-world sensor interfacing, embedded web dashboards, and Access Point (AP) mode networking — all running offline without internet.

PLATFORM

ESP8266 NodeMCU

DOCUMENTATION

Technical Reference

GENERATION DATE

May 2026



Projects Overview

#	Project Title	Relevant SDG(s)	Code & Docs
1	Smart Distance Monitoring System using Ultrasonic Sensor and NodeMCU ESP8266	 SDG 9: Industry, Innovation & Infrastructure  SDG 11: Sustainable Cities & Communities	Smart Distance Monitoring System
2	IoT-Based Ambient Light Intensity Analyzer using NodeMCU	 SDG 7: Affordable & Clean Energy  SDG 13: Climate Action	Ambient Light Intensity Analyzer
3	Smart Soil Moisture Monitoring System for Precision Irrigation	 SDG 2: Zero Hunger  SDG 6: Clean Water & Sanitation	Smart Soil Moisture Monitoring System
4	ClimateSense: IoT Temperature & Humidity Monitoring using DHT11	 SDG 11: Sustainable Cities & Communities  SDG 13: Climate Action	ClimateSense
5	IoT Obstacle Detection and Alert System using IR Sensor	 SDG 3: Good Health & Well-Being  SDG 9: Industry, Innovation & Infrastructure	IoT Obstacle Detection System
6	HeartCare: IoT Heart Beat	 SDG 3: Good Health	HeartCare

#	Project Title	Relevant SDG(s)	Code & Docs
	Monitoring and Visualization System	& Well-Being	
7	Smart Motion Radar with Servo Rotation using Ultrasonic Sensor and NodeMCU	 SDG 9: Industry, Innovation & Infrastructure  SDG 11: Sustainable Cities & Communities	Smart Motion Radar

Development Environment Setup

Follow these steps to set up everything you need to program the ESP8266 NodeMCU board.

Step 1: Install Arduino IDE

1. Go to the official Arduino website: <https://www.arduino.cc/en/software>
2. Download **Arduino IDE 2.x** for your operating system (Windows / macOS / Linux)
3. Run the installer and follow the on-screen instructions
4. Launch Arduino IDE after installation

Note: Arduino IDE 2.x is recommended for the latest features and better performance. Arduino IDE 1.8.x also works if preferred.

Step 2: Install CP210x USB-to-UART Driver

The NodeMCU uses a **CP2102 / CP2104** USB-to-Serial chip. Your computer needs its driver to communicate with the board.

1. Go to the Silicon Labs driver page: <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>

2. Download the **CP210x Universal Windows Driver** (or macOS/Linux version)
3. Extract the ZIP file
4. Run the installer (`CP210xVCPInstaller_x64.exe` for 64-bit Windows)
5. Restart your computer after installation

Verify Installation:

- Plug in your NodeMCU via USB cable
- Open **Device Manager** (Windows) → Look under **Ports (COM & LPT)**
- You should see **Silicon Labs CP210x USB to UART Bridge (COMx)**
- Note the **COM port number** (e.g., COM3) — you'll need this later

Troubleshooting: If the device shows as "Unknown Device" with a yellow warning, try a different USB cable (some are charge-only and don't support data).

Step 3: Install ESP8266 Board Manager

The Arduino IDE doesn't support ESP8266 by default. You need to add it via the Board Manager.

1. Go to **File** → **Preferences** (or `Ctrl + ,`)
2. In the **"Additional Board Manager URLs"** field, paste this URL:

```
https://arduino.esp8266.com/stable/package_esp8266com_index.json
```

If you already have other URLs there, separate them with a comma.

3. Click **OK**
4. Go to **Tools** → **Board** → **Boards Manager...**
5. Search for **"esp8266"**
6. Find **"esp8266 by ESP8266 Community"** and click **Install**
7. Wait for the installation to complete (this may take a few minutes)

Configure Board Settings:

After installation, go to **Tools** and set:

Setting	Value
Board	NodeMCU 1.0 (ESP-12E Module)
Upload Speed	115200
CPU Frequency	80 MHz
Flash Size	4MB (FS:2MB OTA:~1019KB)
Port	Your COM port (e.g., COM3)

Step 4: Install Required Arduino Libraries

Some projects require external libraries. Install them via the **Library Manager**:

1. Go to **Sketch** → **Include Library** → **Manage Libraries...** (or `Ctrl + Shift + I`)
2. Search and install each library listed below:

Required Libraries

Library	Used By	How to Install
DHT11	ClimateSense	Search <code>DHT11</code> in Library Manager → Install by Dhruba Saha
Servo	Smart Motion Radar	Usually pre-installed. If not, search <code>Servo</code> → Install by Arduino
ESP8266WiFi	All projects	Included with ESP8266 Board Manager (no separate install needed)
ESP8266WebServer	All web projects	Included with ESP8266 Board Manager (no separate install needed)

Important: The `ESP8266WiFi` and `ESP8266WebServer` libraries are automatically installed when you install the ESP8266 Board Manager in Step 3. Do NOT try to install them separately.

Step 5: Upload Your First Sketch

1. Open any `.ino` file from this repository (e.g., `HeartCare/HeartCare.ino`)
 2. Connect your NodeMCU via USB
 3. Select the correct **Board** and **Port** from the **Tools** menu
 4. Click the **Upload** button (→ arrow) or press `Ctrl + U`
 5. Wait for the upload to complete — you should see `Done uploading.` in the console
 6. Open **Serial Monitor** (`Ctrl + Shift + M`) and set baud rate to **9600** to see debug output
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How to Use the Web Dashboards

Each project acts as an offline, standalone **Wi-Fi Access Point** (AP) that hosts its own dashboard.

1. **Upload** the sketch to the NodeMCU board.
2. On your phone, tablet, or laptop, open your **Wi-Fi Settings**.
3. Scan for networks and connect to the specific project's network (SSID):

#	Project Name	Wi-Fi Access Point SSID	Password	IP Address
1	Smart Distance Monitoring System	MeasureTool_AP	12345678	192.168.4.1
2	Ambient Light Intensity Analyzer	LuxMeter_AP	12345678	192.168.4.1
3	Smart Soil Moisture Monitoring System	AgriMonitor_AP	12345678	192.168.4.1
4	ClimateSense	ClimateSense_AP	12345678	192.168.4.1
5	IoT Obstacle Detection & Alert System	ObstacleAlert_AP	12345678	192.168.4.1
6	HeartCare	HeartCare_AP	12345678	192.168.4.1
7	Smart Motion Radar	SmartRadar_AP	12345678	192.168.4.1

4. When prompted, enter the password: **12345678**

5. Open your web browser and navigate to: **http://192.168.4.1**

6. The interactive web dashboard will load with live telemetry and visualizations.

No Internet Connection Required! The ESP8266 acts as a self-contained local server. Everything runs directly on the hardware.

Common Pin Connections

NodeMCU Pin	GPIO	Common Use in Projects
A0	ADC0	Analog sensors (Pulse, Soil, TEMT6000)
D2	GPIO4	LED indicator / Servo signal
D4	GPIO2	DHT11 data pin
D5	GPIO14	HC-SR04 Trigger
D6	GPIO12	HC-SR04 Echo (use voltage divider!)
D7	GPIO13	Buzzer

 **Voltage Divider Required:** The HC-SR04 Echo pin outputs 5V, but ESP8266 GPIO pins are **3.3V tolerant only**. Use a simple resistor voltage divider (1k Ω + 2k Ω) to step down the voltage.

License

These projects are created for educational purposes as part of KASCC 2026. Feel free to use, modify, and learn from them.